AI and the Production of Ideas

Annie Liang
(Northwestern and OpenAI)

Tylor Wu
(Northwestern)

SMART Workshop II

Introduction

- Many expect artificial intelligence (AI) to substantially accelerate research, potentially leading to faster economic growth (Aghion et al, 2019; Jones and Tonetti, 2026)
- Evidence that AI raises **individual research productivity**, lowering costs and improving output quality (Codreanu et al., 2026; Bubeck et al., 2025).
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- But individual gains need not translate into **aggregate progress**.
- Some studies have found AI to improve outputs but **increase dependence** across them (Doshi and Hauser 24, Meincke et al., 25), and that LLM outputs are often correlated with one another (Ballestero et al., 26)
- When progress depends on the **best ideas discovered**, this dependence is harmful

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Our question: If AI increases research volume but makes ideas more dependent, what happens to the scientific frontier?

This talk

- We propose a minimal model to study this question
- The answer crucially depends on our assumptions about the tail of the idea distribution

Model

Model

Each researcher $i = 1,\ldots,n$ chooses:

- an **AI-reliance parameter** $\alpha_i \in [0,1]$
  - $\alpha_i = 0$ corresponds to **pure human generation**
  - $\alpha_i = 1$ corresponds to **pure AI generation**
- a **number of ideas** $m_i \in \mathbb{N}_+$ to produce

Researcher i's k-th idea has **random** quality

$$Y_{ik} = \mu_{\alpha_i} + \lambda_{\alpha_i} Z + s_{\alpha_i} \varepsilon_{ik}$$

mean

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common shock

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ε_{ik}: idiosyncratic shock

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mean zero and independent

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- **better** up to a point: μ_{α_i} increases up to $\overline{\alpha} \in (0,1]$, then decreases

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Special Case: Gaussian Ideas

Idea qualities $(Y_{1,1}, \ldots, Y_{1,m_1}; \ldots; Y_{n,1}, \ldots, Y_{n,m_n})$ are jointly Gaussian with

$$\mathbb{E}[Y_{ik}] = \mu(\alpha_i) \qquad \mathrm{Cov}(Y_{ik}, Y_{jl}) = \begin{cases} 1 & \text{if } (i,k) = (j,l) \\ \sqrt{\rho(\alpha_i)\rho(\alpha_j)} & \text{otherwise} \end{cases}$$

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Corresponds to $Y_{ik} = \mu_{\alpha_i} + \lambda_{\alpha_i} Z + s_{\alpha_i} \varepsilon_{ik}$ with

- $\lambda_{\alpha_i} = \sqrt{\rho(\alpha_i)}$ and $s_{\alpha_i} = \sqrt{1 - \rho(\alpha_i)}$
- $Z \perp \varepsilon_{ik}$ and both standard normal

Research Frontier

For any action profile $a = ((\alpha_1, m_1), \ldots, (\alpha_n, m_n))$, let $Y_a^{(j)}$ denote the j-th highest quality among all ideas produced under the profile a.

Definition: The research frontier at profile a is

$$F_n(a) = \mathbb{E}\left[\sum_{j=1}^{L} w_j Y_a^{(j)}\right]$$

for some $L < n$ and weights w_k summing to 1.

- Special case is simply the best idea (Kortum, 1997; Jones, 2023)

Researcher Payoffs

Each researcher i's utility function consists of two parts:

- A **professional** payoff capturing costs/benefits to producing ideas

- A **social** payoff capturing the benefits they receive from research in their role as citizens of the broader community
    - Everyone benefits when someone finds a cure to a disease

Professional Payoffs

$$\pi(\alpha_i, m_i) = \sum_{k=1}^{m_i} r(Y_{ik}) - c_{\alpha_i}(m_i)$$

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PRIVATE REWARD

sum of individual rewards per idea

Assumption: The reward r is increasing in the idea's quality

Professional Payoffs

PRIVATE COST

$$\pi(\alpha_i, m_i) = \sum_{k=1}^{m_i} r(Y_{ik}) - \boxed{c_{\alpha_i}(m_i)}$$

cost of producing m_i ideas at reliance α_i

Assumption: The reward r is increasing in the idea's quality

Assumption: The cost c is increasing in m_i (for fixed α_i) and decreasing in α_i (for fixed m_i)

Assumption: $\inf_{\alpha \in [0,1]} (c_\alpha(m+1) - c_\alpha(m)) \to 0$, i.e., marginal costs diverge

Full Utility Function

Each researcher i's payoff at action profile a is

$$U_i(a) = \pi(\alpha_i, m_i) + \gamma \cdot F_n(a)$$

PROFESSIONAL PAYOFF

PAYOFF AS CITIZEN

$\gamma > 0$ determines the weight placed on the social benefit

Equilibrium and Welfare

- Let Γ_n^{AI} denote the game with
  - Agents $i \in \{1,\ldots,n\}$
  - Action sets $A_i = [0,1] \times \mathbb{N}_+$
  - Payoffs U_i

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Proposition: Equilibrium exists in both games Γ_n^{AI} and Γ_n^0.

Our results subsequently compare welfare in equilibrium across these games.

Plan for Results

- Gaussian case
- Heavy-tailed case
- Intuition for difference

Results

(i) Gaussian Case

$$Y_{ik} = \mu_{\alpha_i} + \sqrt{\rho(\alpha_i)}\, Z + \sqrt{1 - \rho(\alpha_i)}\, \varepsilon_{ik}$$

$Z \perp \varepsilon_{ik}$ and both standard normal

Professionally-Optimal Actions

Recall the professional payoff:

$$\pi(\alpha_i, m_i) = \sum_{k=1}^{m_i} r(Y_{ik}) - c(m_i, \alpha_i)$$

Definition: Let (α^*, m^*) denote the maximizer of $\pi(\alpha, m)$, and let $(0,m_0)$ denote the maximizer of $\pi(0,m)$

- Let's just assume unique, qualitative conclusions hold without this

These are the actions that maximize the researcher's professional payoffs in the games with and without AI.

Research Frontier for Gaussian Ideas

Proposition: Let (σ_n^{AI}) and (σ_n^0) respectively denote sequences of Nash equilibria of the game with AI, Γ_n^{AI}, and the game without AI, Γ_n^0. Then:

$$\frac{E_{\sigma_n^{AI}}[F_n(a)]}{E_{\sigma_n^0}[F_n(a)]} \longrightarrow \sqrt{\frac{1-\rho(\alpha^*)}{1-\rho(0)}} < 1.$$

- In large populations, AI lowers the scientific frontier by a strict fraction of what it would have been without AI.
- The size of that fraction is determined by the increase in correlation from $\rho(0)$ to $\rho(\alpha^*)$
- Although AI expands each researcher's production frontier, it worsens the best ideas generated in the community

Welfare for Gaussian Ideas

Proposition: Let (σ_n^{AI}) and (σ_n^0) respectively denote sequences of NE of the game with AI, Γ_n^{AI}, and the game without, Γ_n^0. For all sufficiently large n,

$$E_{\sigma_n^{AI}}[U_i(a)] < E_{\sigma_n^0}[U_i(a)] \quad \text{for all agents } i = 1,\ldots,n$$

- Access to AI makes the community worse off **in the strong sense of Pareto dominance**: every agent's utility is lower

Proof Intuition (1/3)

In equilibrium, agents' actions converge to (α^*, m^*), the maximizer of their professional payoff

- Classic collective action problem: agents do not internalize the externality imposed by correlating their ideas with those of others
- As the population grows large, any given agent becomes increasingly unlikely to produce one of the best ideas.
- Not optimal to sacrifice professional payoff to improve the frontier by (in expectation) a vanishing amount
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  - Hence too much AI reliance and too much correlation in equilibrium

Different from collective action problems, there is a potentially offsetting factor: researchers can produce more and better ideas at lower cost.

One might expect this to compensate for loss in variation. We show it does not.

Proof Intuition (2/2)

To see why, consider a symmetric equilibrium in which all agents choose reliance α_n and quantity m_n.

The best idea has expected quality

$$E\left[Y_a^{(1)}\right] = E\left[\mu(\alpha_n) + \sqrt{\rho(\alpha_n)}Z + \sqrt{1-\rho(\alpha_n)}\varepsilon_a^{(1)}\right]$$

largest idiosyncratic shock

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$E[\mu(\alpha_n)]$ $\downarrow$ $E[\mu(\alpha^*)]$ uniformly bounded in a

$\sqrt{\rho(\alpha_n)}E[Z]$ $=$ 0

$E\left[\varepsilon_a^{(1)}\right]$ $\downarrow$ $\sqrt{2\log(nm_n)}$

Proof Intuition (2/3)

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The best idea has expected quality

$$E\left[Y_a^{(1)}\right] = E[\mu(\alpha_n)] + \sqrt{\rho(\alpha_n)}E[Z] + \sqrt{1-\rho(\alpha_n)}E\left[\varepsilon_a^{(1)}\right]$$

$E[\mu(\alpha_n)] \to E[\mu(\alpha^*)]$ (uniformly bounded in a)

$E[Z] = 0$

$E\left[\varepsilon_a^{(1)}\right] \to \sqrt{2\log(nm_n)}$

Proof Intuition (3/3)

- So the expected frontier is eventually approximately

$$\sqrt{1-\rho(\alpha_n)}\sqrt{2\log(nm_n)}$$

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- Now recall that $\alpha_n \to \alpha^*$ and $m_n \to m^*$ (the privately optimal choice).

- The additional quantity contribution per person is eventually bounded. It enters **additively** into $\sqrt{2(\log n + \log m^*)}$ and is of vanishing importance.

- The reduction due to correlation ($\sqrt{1-\rho(\alpha^*)}$) enters multiplicatively, and its effect does not go away even as n grows large.

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- Taking ratios yields the limiting constant $\sqrt{\frac{1-\rho(\alpha^*)}{1-\rho(0)}}$ (Prop 1)

- Argument extends when (1) we consider a weighted average of the best ideas, (2) we consider possibly asymmetric/mixed equilibria

Takeaways

- When ideas are normally distributed, a (bounded) increase in research volume is **not** enough to compensate for increased correlation across ideas
- Note that the ranking of welfare and of the research frontier holds for any increasing ρ (that is, even a little more correlation can be very harmful!)
- But the size of loss depends on the shape of ρ
  - e.g., when ρ is flatter (less sensitive to AI reliance) then the limiting frontier ratio $\sqrt{\frac{1-\rho(\alpha^*)}{1-\rho(0)}}$ will be closer to 1
- Next show that this strong negative conclusion relied on the assumption of Gaussian ideas

Results

(ii) Heavy-Tailed Case

Heavy-Tailed Case

Recall our general model in which researcher i's k-th idea has quality

$$Y_{ik} = \mu_{\alpha_i} + \lambda_{\alpha_i} Z + s_{\alpha_i} \varepsilon_{ik}$$

Now assume this idiosyncratic shock has a heavy tail

Assumption: $P(\varepsilon_{ik} > t) \sim Dt^{-\tau}$ for $\tau > 1$ as $t \to \infty$ (e.g., Pareto distribution)

Returns to Scale Index

Definition: The **returns to scale** index is

$$\Theta := \frac{s_{\alpha^*}}{s_0} \left(\frac{m^*}{m_0} \right)^{1/\tau}$$

This quantifies the extent to which AI increases quantity relative to the no-AI benchmark, where we take into account:

- the effectiveness of new draws at generating breakthroughs (determined by the tail parameter τ)
- the redundancy of these draws (determined by the weight on the idiosyncratic component)

Research Frontier for Heavy-Tailed Ideas

Theorem: Let (σ_n^{AI}) and (σ_n^0) respectively denote sequences of Nash equilibria of the game with AI, Γ_n^{AI}, and the game without AI, Γ_n^0. Then:

$$\frac{E_{\sigma_n^{AI}}[F_n(a)]}{E_{\sigma_n^0}[F_n(a)]} \longrightarrow \Theta$$

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$$\frac{E_{\sigma_n^{AI}}[F_n(a)]}{E_{\sigma_n^0}[F_n(a)]} \longrightarrow \Theta$$

Moreover,

(a) If $\Theta > 1$ then for all sufficiently large n

$$E_{\sigma_n^{AI}}[U_i(a)] > E_{\sigma_n^0}[U_i(a)] \quad \forall \text{ agents } i = 1,\ldots,n$$

(b) If $\Theta < 1$ then for all sufficiently large n

$$E_{\sigma_n^{AI}}[U_i(a)] < E_{\sigma_n^0}[U_i(a)] \quad \forall \text{ agents } i = 1,\ldots,n$$

Intuition

- The incentive problem remains (researchers still do not internalize the effect of their choices on the frontier)

- But when the idea distribution is heavy-tailed, expansion in quantity has a first-order impact on the expected order statistics
  - e.g., doubling the number of ideas (asymptotically) improves the frontier from $\sqrt{2\log(N)}$ to $\sqrt{2\log(2N)}$ when ideas are Gaussian
  - but it (asymptotically) improves the frontier from $(N)^{1/\tau}$ to $(2N)^{1/\tau}$ when ideas are heavy-tailed

- This latter improvement does not vanish as the number of ideas grows large, and can compensate for the increased dependence across ideas

Takeaways

- Improvement to individual research processes **need not** translate into aggregate improvement in the research frontier
- Depends on:
    - Whether and how much AI reliance introduces dependence across ideas
        - Tail properties of the idea distribution
- Both of the above are quantities that can be **measured** and **changed**
    - AI labs can seek to reduce dependence across outputs
    - We can modify research rewards to incentivize diversity in outputs
- Need to go beyond individual-level measures of AI's impact on science

Extensions/Variations

Extension 1: Other Utility Functions

	Scientific value	
Professional reward	Best ideas overall	Sum of ideas
Sum across ideas	**(A)**	**(B)**
Best ideas overall	**(C)**	**(D)**

(A) is this paper

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	Scientific value	
Professional reward	Best ideas overall	Sum of ideas
Sum across ideas	**(A)**	**(B)**
Best ideas overall	**(C)**	**(D)**

(B): Additive scientific value removes the cost of correlation, leaving a familiar public-goods problem

- Researchers still do not fully internalize the benefits their research provides to others.
- But AI can improve the frontier/welfare if it increases quality-adjusted research output.

Extension 1: Other Utility Functions

	Scientific value	
Professional reward	Best ideas overall	Sum of ideas
Sum across ideas	**(A)**	**(B)**
Best ideas overall	**(C)**	**(D)**

(C): Individual research efforts increasingly unlikely to matter

- Researchers minimize costs by fully relying on AI (we don't allow exit)
- Obtain the stark negative result (frontier and welfare are both lower with AI) under both sets of distributional assumptions

Extension 1: Other Utility Functions

	Scientific value	
Professional reward	Best ideas overall	Sum of ideas
Sum across ideas	**(A)**	**(B)**
Best ideas overall	**(C)**	**(D)**

(D): Useful ideas can benefit society without earning professional rewards — underpowered incentives to produce ideas (and hence also a weird case)

Extension 1: Other Utility Functions

	Scientific value	
Professional reward	Best ideas overall	Sum of ideas
Sum across ideas	**(A)**	**(B)**
Best ideas overall	**(C)**	**(D)**

Direction for future work: more sophisticated reward systems.

Extension 2: Macroeconomic Growth

- So far have considered a static game
- A line in macro models research as sampling for ideas, with the best idea determining the technological frontier (Kortum, 1997; Jones, 2023)
- Our model fits directly into this framework and answers a question left open in Jones (2023) about the impact of correlation in this sampling
- Dynamic extension:
  - The population grows over time
  - AI research costs and dependence also evolve over time
- Our frontier comparisons now map into comparisons of economic growth rates

Conclusion

- We model research in a world with AI, focusing on the impact of increased similarity across ideas
- Three points:
  - Individual productivity gains need not translate into collective progress
  - The tail of idea quality is crucial to the tradeoff
  - Evaluation and rewards should correctly incentive scientific contribution

Thank You